

# Microsoft Defender for Endpoint Detection and Incident Investigation Lab

Eric Johnson

## Executive Summary

This project demonstrates the deployment, configuration, validation, and investigation capabilities of Microsoft Defender for Endpoint (MDE) within a Microsoft Defender XDR environment. A Windows 11 endpoint was onboarded to Microsoft Defender for Endpoint; endpoint telemetry was validated, and a Microsoft detection test was executed to simulate suspicious PowerShell activity. The generated alerts were automatically correlated into a security incident, which was then investigated using Device Timeline analysis and the Microsoft Defender XDR Attack Story Graph.

## Introduction

Microsoft Defender for Endpoint is a cloud-native endpoint detection and response (EDR) platform designed to provide endpoint visibility, threat detection, automated investigation, and incident response capabilities. The objective of this project was to deploy Microsoft Defender for Endpoint within a home lab environment, validate endpoint telemetry collection, generate a controlled security event, and investigate the resulting incident using Microsoft Defender XDR.

## Lab Environment

| Component               | Configuration                       |
|-------------------------|-------------------------------------|
| Host Operating System   | Windows 11 Pro 25H2                 |
| Security Platform       | Microsoft Defender XDR              |
| Endpoint Security       | Microsoft Defender for Endpoint P2  |
| Virtualization Platform | VMware Workstation                  |
| Device Name             | djm                                 |
| Detection Method        | Microsoft PowerShell Detection Test |
| Investigation Platform  | Microsoft Defender XDR              |

## Phase 1: Endpoint Onboarding

The Windows 11 endpoint was onboarded into Microsoft Defender for Endpoint using the Local Script onboarding method available within Microsoft Defender XDR. The onboarding package was generated through the Defender portal and executed with administrative privileges on the endpoint.

Successful onboarding was validated by confirming:

- Sensor Health = Active
- Onboarding Status = Onboarded
- Device visibility within Microsoft Defender XDR
- Endpoint telemetry generation

Figure 1 – Microsoft Defender for Endpoint Device Inventory Showing Successful Endpoint Onboarding

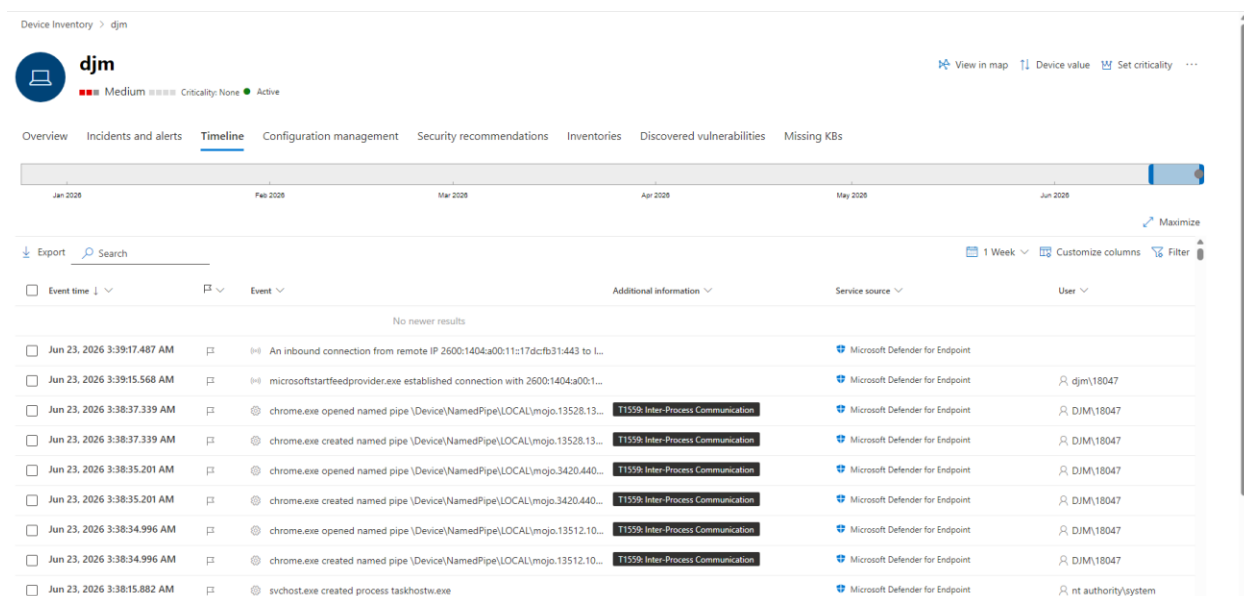
| Name                | IP                | Criticality level | Device category      | Device type | Domain      | Device AAD id                        | Risk level | Exposure level    | OS platform       | OS version      | Sensor health state |
|---------------------|-------------------|-------------------|----------------------|-------------|-------------|--------------------------------------|------------|-------------------|-------------------|-----------------|---------------------|
| djm                 | 192.168.5.1       |                   | Computers and Mo...  | Workstation | Workgroup   | 15e3c90c-f54a-4188-8ac3-9000c8f0c1ac | Medium     | No data available | Windows 11        | 25H2            | Active              |
| Sensor health state | Onboarding status | Discovery sources | Last device update   | Tags        | Device role | Security operations                  | Managed by | Managed by status | Mitigation status | Cloud platforms |                     |
| Active              | Onboarded         |                   | Jun 23, 2026 2:33 AM |             |             | Full                                 | Unknown    | Unknown           |                   |                 |                     |

Phase 2: Telemetry Validation

Following onboarding, endpoint telemetry was validated using the Microsoft Defender Device Timeline. Telemetry collected included process creation events, file creation activity, DNS requests, and network communications.

These events confirmed successful communication between the endpoint and Microsoft Defender for Endpoint.

Figure 2 – Device Timeline Telemetry



## Phase 3: Detection Validation

To validate Microsoft Defender's detection capabilities, Microsoft's built-in detection test was executed using a PowerShell command provided within the onboarding portal.

The test simulated suspicious PowerShell activity commonly associated with attacker tradecraft and generated a security alert within Microsoft Defender XDR.

## Detection Generated

| Detection                          | Severity |
|------------------------------------|----------|
| Suspicious PowerShell Command Line | Medium   |

## MITRE ATT&CK Mapping

| Technique ID | Technique         |
|--------------|-------------------|
| T1059.001    | PowerShell        |
| T1057        | Process Discovery |

Figure 3 – Suspicious PowerShell Command Line Alert

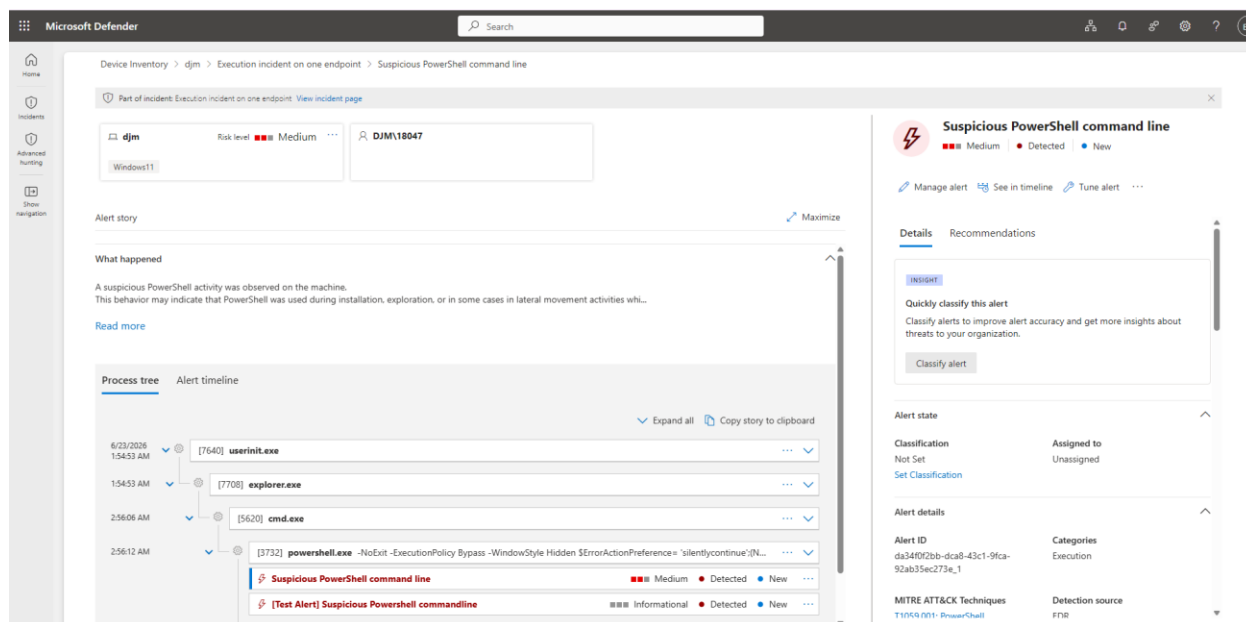


Figure 3. Microsoft Defender for Endpoint generated a Suspicious PowerShell Command Line alert following execution of a controlled detection validation test. The alert details page displays the associated process tree, affected endpoint, user context, severity classification, and MITRE ATT&CK mapping.

## Phase 4: Incident Investigation

Microsoft Defender XDR automatically correlated multiple related alerts into a single incident.

The incident was categorized as an Execution incident and contained two correlated PowerShell-related alerts associated with the endpoint.

### Incident Details

| Field           | Value                              |
|-----------------|------------------------------------|
| Incident Name   | Execution incident on one endpoint |
| Severity        | Medium                             |
| Category        | Execution                          |
| Alerts          | 2                                  |
| Impacted Device | djm                                |
| Status          | Active                             |

Figure 4 – Incident Summary

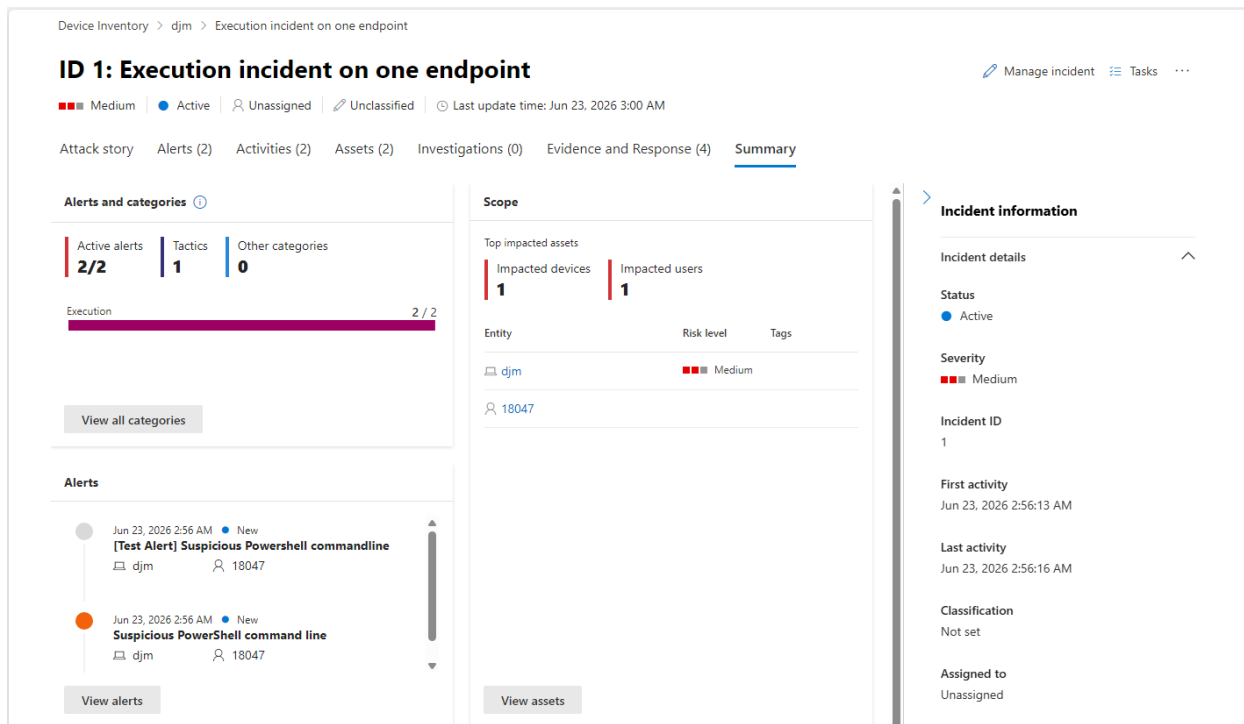


Figure 4. Microsoft Defender XDR incident summary showing automated correlation of multiple PowerShell-related alerts into a single security incident. The incident identifies the affected endpoint, user account, alert count, severity, and investigation scope.

## Phase 5: Attack Story Analysis

The Microsoft Defender XDR Attack Story Graph was used to visualize relationships between the user, endpoint, PowerShell process, and associated network activity.

The graph automatically mapped affected entities and displayed the attack path associated with the suspicious PowerShell activity.

Figure 5 – Microsoft Defender XDR Attack Story Graph

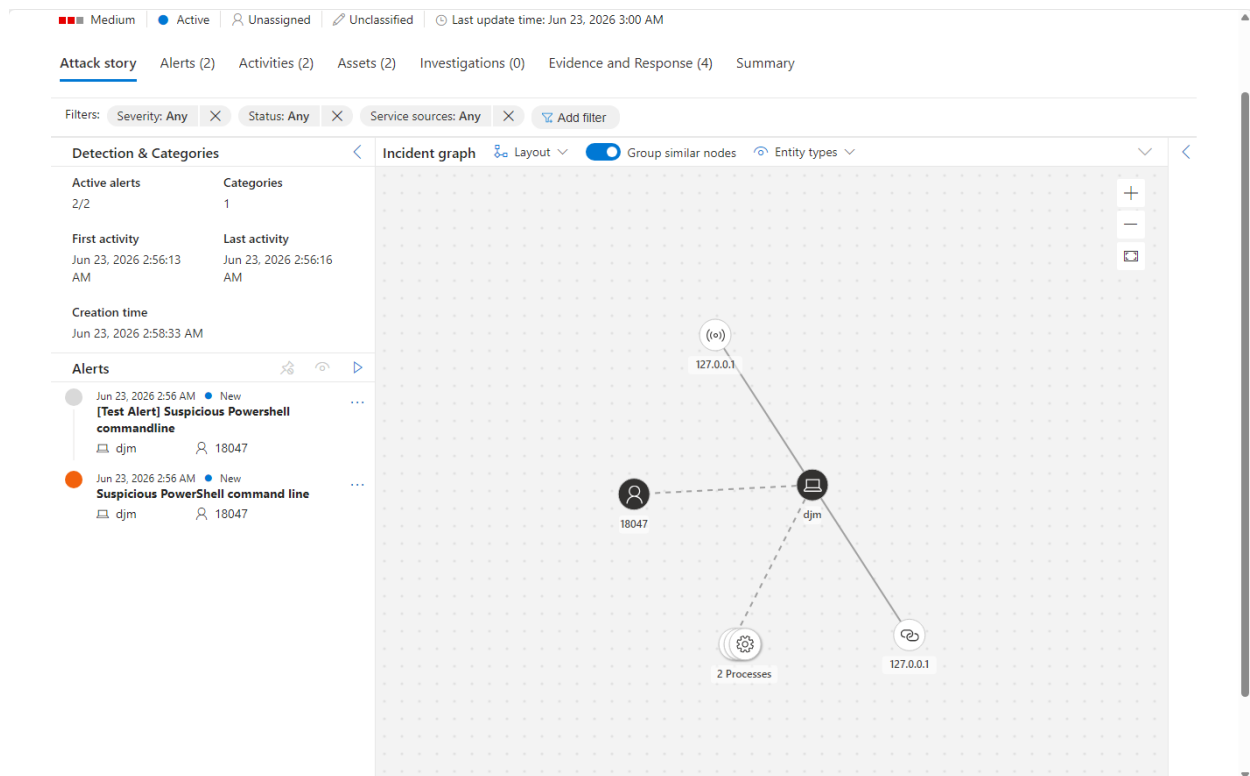


Figure 5. Microsoft Defender XDR Attack Story Graph visualizing relationships between the affected endpoint, user account, processes, IP addresses, and associated PowerShell detections during incident investigation.

## Key Findings

- Microsoft Defender for Endpoint successfully onboarded and collected endpoint telemetry.
- Detection validation generated a Medium severity Suspicious PowerShell Command Line alert.
- Microsoft Defender XDR automatically correlated related alerts into a single incident.
- Device Timeline and Attack Story analysis provided visibility into process execution and affected entities.
- No malicious activity was identified; the alert resulted from a controlled detection validation test.

## Conclusion

This project demonstrated the deployment, validation, and investigation capabilities of Microsoft Defender for Endpoint within a Microsoft Defender XDR environment. The investigation validated endpoint telemetry collection, alert generation, incident correlation, and Attack Story visualization. Through the use of Device Timeline analysis and incident investigation workflows, the project reinforced practical Security Operations Center (SOC) analyst skills related to alert triage, endpoint investigation, and threat hunting.